

## EAST LONDON, South Africa

WMO: 688580

Lat: 33.0358S

Lon: 27.8161E

Elev: 116

StdP: 99.94

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 8.0        | 9.0 | -2.5                          | 3.1 | 18.0 | -0.6 | 3.6 | 17.1 | 13.3                  | 15.7 | 12.1 | 15.4 | 3.5                   | 330  | 0.445 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 6.8                    | 30.8            | 20.0 | 28.9 | 20.5 | 27.3 | 20.7 | 23.8                | 27.5 | 23.1 | 26.3 | 22.5 | 25.5 | 4.7                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.8                            | 17.8 | 25.7 | 22.1 | 17.0 | 24.9 | 21.3 | 16.2 | 24.3 | 72.0          | 27.7 | 69.2 | 26.5 | 66.9 | 25.4 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |     |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|-----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |     |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |     |
| (4) 12.1          | 10.6 | 9.5 | DB                         | 5.1 | 35.7               | 1.3 | 2.5                                                | 4.1 | 37.5       | 3.4 | 39.0       | 2.7 | 40.4       | 1.7 | 42.2 | (4) |
| (5)               |      |     | WB                         | 3.2 | 25.6               | 0.9 | 1.3                                                | 2.5 | 26.5       | 2.0 | 27.2       | 1.4 | 28.0       | 0.8 | 28.9 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.8  | 21.9  | 22.4  | 21.3  | 19.5  | 18.1  | 16.3  | 15.9  | 16.3  | 16.9  | 18.0  | 19.1  | 20.8 | (6)  |
| (7)  |                                                                                  | DBStd       | 3.26  | 2.01  | 1.96  | 2.07  | 2.48  | 2.61  | 2.72  | 2.91  | 2.78  | 2.31  | 2.31  | 2.21  | 2.13 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 403   | 1     | 1     | 2     | 14    | 36    | 74    | 89    | 78    | 55    | 34    | 16    | 3    | (9)  |
| (10) |                                                                                  | CDD10.0     | 3230  | 369   | 346   | 350   | 287   | 250   | 189   | 183   | 194   | 206   | 247   | 273   | 335  | (10) |
| (11) |                                                                                  | CDD18.3     | 592   | 112   | 113   | 94    | 50    | 27    | 14    | 14    | 13    | 11    | 23    | 39    | 80   | (11) |
| (12) |                                                                                  | CDH23.3     | 2417  | 416   | 462   | 343   | 193   | 156   | 109   | 117   | 115   | 57    | 81    | 109   | 258  | (12) |
| (13) | CDH26.7                                                                          | 490         | 64    | 74    | 61    | 45    | 47    | 30    | 34    | 43    | 13    | 17    | 20    | 41    | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 4.7   | 4.8   | 4.7   | 4.5   | 4.2   | 4.4   | 4.4   | 4.8   | 5.0   | 5.2   | 5.2   | 5.0   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 863   | 77    | 84    | 98    | 78    | 46    | 36    | 43    | 48    | 71    | 97    | 107   | 73   | (15) |
| (16) |                                                                                  | PrecMax     | 1350  | 230   | 229   | 600   | 325   | 159   | 196   | 221   | 224   | 258   | 314   | 386   | 243  | (16) |
| (17) |                                                                                  | PrecMin     | 564   | 14    | 17    | 5     | 7     | 0     | 0     | 0     | 0     | 2     | 5     | 12    | 7    | (17) |
| (18) |                                                                                  | PrecStd     | 194   | 47    | 51    | 92    | 64    | 43    | 45    | 57    | 44    | 54    | 72    | 79    | 51   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 31.1  | 31.1  | 31.0  | 31.3  | 31.2  | 30.0  | 30.2  | 31.9  | 28.8  | 28.9  | 28.9  | 30.2 | (19) |
| (20) |                                                                                  |             | MCWB  | 23.4  | 24.1  | 22.7  | 20.8  | 17.9  | 14.9  | 15.2  | 15.8  | 17.6  | 18.9  | 21.1  | 22.3 | (20) |
| (21) |                                                                                  | 2%          | DB    | 28.2  | 28.8  | 28.2  | 27.8  | 27.1  | 26.9  | 26.8  | 26.2  | 24.8  | 25.2  | 25.9  | 27.5 | (21) |
| (22) |                                                                                  |             | MCWB  | 23.0  | 23.3  | 22.1  | 20.7  | 17.8  | 14.6  | 14.5  | 15.4  | 17.4  | 19.2  | 20.5  | 21.9 | (22) |
| (23) |                                                                                  | 5%          | DB    | 26.9  | 27.2  | 26.5  | 25.4  | 24.3  | 23.4  | 23.2  | 23.0  | 22.8  | 23.2  | 24.2  | 26.0 | (23) |
| (24) |                                                                                  |             | MCWB  | 22.2  | 22.5  | 21.4  | 19.8  | 16.7  | 14.0  | 13.9  | 15.1  | 16.9  | 18.4  | 19.6  | 21.3 | (24) |
| (25) |                                                                                  | 10%         | DB    | 25.6  | 26.1  | 25.1  | 23.9  | 22.7  | 21.2  | 21.0  | 21.0  | 21.1  | 21.9  | 23.0  | 24.8 | (25) |
| (26) |                                                                                  |             | MCWB  | 21.5  | 21.8  | 20.7  | 19.2  | 16.5  | 13.8  | 13.4  | 14.7  | 16.3  | 17.6  | 18.8  | 20.6 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 24.7  | 25.1  | 24.1  | 22.7  | 20.7  | 18.1  | 17.4  | 18.9  | 20.1  | 21.1  | 22.2  | 23.7 | (27) |
| (28) |                                                                                  |             | MCDWB | 28.5  | 29.3  | 28.1  | 27.8  | 26.1  | 22.3  | 22.2  | 23.4  | 23.8  | 25.7  | 26.3  | 27.7 | (28) |
| (29) |                                                                                  | 2%          | WB    | 23.5  | 24.0  | 22.9  | 21.5  | 19.3  | 16.7  | 16.4  | 17.7  | 18.7  | 19.9  | 21.2  | 22.7 | (29) |
| (30) |                                                                                  |             | MCDWB | 26.9  | 27.7  | 26.5  | 25.7  | 23.7  | 20.9  | 21.0  | 21.4  | 21.9  | 23.6  | 24.4  | 25.7 | (30) |
| (31) |                                                                                  | 5%          | WB    | 22.8  | 23.2  | 22.2  | 20.7  | 18.3  | 15.8  | 15.6  | 16.9  | 18.0  | 19.1  | 20.4  | 22.0 | (31) |
| (32) |                                                                                  |             | MCDWB | 25.7  | 26.2  | 25.4  | 24.3  | 22.2  | 20.5  | 20.3  | 20.5  | 21.2  | 22.2  | 23.2  | 24.8 | (32) |
| (33) |                                                                                  | 10%         | WB    | 22.1  | 22.5  | 21.4  | 19.8  | 17.4  | 15.0  | 14.8  | 16.1  | 17.1  | 18.4  | 19.7  | 21.2 | (33) |
| (34) |                                                                                  |             | MCDWB | 24.9  | 25.2  | 24.3  | 22.8  | 21.2  | 19.9  | 19.5  | 19.5  | 20.2  | 21.1  | 22.2  | 23.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.7   | 6.8   | 7.2   | 8.0   | 9.5   | 10.4  | 10.3  | 9.5   | 8.3   | 7.4   | 7.0   | 6.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 8.7   | 8.6   | 9.5   | 10.6  | 12.9  | 13.7  | 14.6  | 13.8  | 11.4  | 9.9   | 9.0   | 8.9  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.7   | 4.6   | 4.9   | 5.6   | 6.7   | 6.6   | 7.1   | 6.9   | 6.2   | 5.5   | 5.3   | 4.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 7.6   | 7.9   | 8.5   | 9.2   | 10.7  | 11.2  | 11.3  | 10.5  | 9.5   | 8.8   | 8.1   | 7.8  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.5   | 4.6   | 4.8   | 5.3   | 6.4   | 6.9   | 6.8   | 6.3   | 6.0   | 5.2   | 5.0   | 4.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.406 | 0.406 | 0.390 | 0.362 | 0.343 | 0.327 | 0.330 | 0.382 | 0.409 | 0.418 | 0.398 | 0.401 | (40) |      |
| (41) |                                                                                  | taud        | 2.424 | 2.434 | 2.457 | 2.503 | 2.525 | 2.536 | 2.521 | 2.373 | 2.296 | 2.314 | 2.409 | 2.429 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 933   | 912   | 890   | 861   | 827   | 816   | 831   | 825   | 856   | 890   | 935   | 941   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 123   | 118   | 108   | 92    | 80    | 74    | 79    | 103   | 125   | 132   | 124   | 123   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.37  | 5.87  | 4.95  | 3.94  | 3.13  | 2.80  | 3.00  | 3.69  | 4.65  | 5.39  | 6.19  | 6.50  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.40  | 0.35  | 0.39  | 0.23  | 0.14  | 0.15  | 0.18  | 0.22  | 0.33  | 0.46  | 0.38  | 0.41  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.24 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (7 neighbors) | +0.24 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page